101. (c) $\underset{\text{Methane}}{CH_4} + 2O_2 \rightarrow CO_2 + 2H_2O$

$\underset{\text{Ethene}}{C_2H_4} + 3O_2 \rightarrow 2CO_2 + 2H_2O$

All hydrocarbons saturated or unsaturated on complete combustion always produce CO_2 and H_2O.

102. (a) Free rotation around carbon – carbon bond takes place easily in alkanes. Now ethane and hexachloroethane both are alkanes. But in hexachloroethane bulky chlorine atom hinders the rotation. Therefore least hindered rotation takes place in ethane.

103. (a) Hydrocarbons on complete oxidation produce CO_2 and water

$CH_3 - CH_3 + 3\frac{1}{2}O_2 \rightarrow 2CO_2 + 3H_2O$

104. (b) $\underset{\text{Decane}}{C_{10}H_{22}} \xrightarrow[\text{Cracking}]{900\ K} \underset{\text{Butane}}{C_4H_8} + \underset{\text{Hexane}}{C_6H_{14}}$

105. **(d) Four** — It contains four types of carbon atoms: primary (1°), secondary (2°), tertiary (3°), and quaternary (4°) carbons.

106. (b) $CH_3 - \underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}} - CH_3$

Replaceable hydrogen atoms are present only on 4 primary carbon atoms. Hence, it gives only are monochloro Substituted product.

107. (d) $\underset{\text{Sod. acetate}}{CH_3COONa} + NaOH \xrightarrow{CaO} CH_4 + Na_2CO_3$

$\underset{\text{Iodomethan e}}{CH_3I} + 2H \xrightarrow[Reduction]{} CH_4 + HI$

$\underset{\text{Aluminiumcarbide}}{Al_4C_3} + 6H_2O \rightarrow 3CH_4 + 2Al_2O_3$

108. (c) Acetylene reacts with ammonical cuprous chloride to form red ppt. of copper acetylide while methane and ethylene do not react (since they do not have acidic hydrogen) They come out from the bottle

$CH \equiv CH + 2[Cu(NH_3)_2]OH \rightarrow Cu - C \equiv C - Cu + 4NH_3 + 2H_2O$

$CH_4 + \text{Ammonical } Cu_2Cl_2 \rightarrow \text{No reaction}$

$C_2H_4 + \text{Ammonical } Cu_2Cl_2 \rightarrow \text{No reaction}$

109. **(d)** $C_{20}H_{42}$ — Higher alkanes (around C_{18} and above) are solids at room temperature; paraffin wax consists of long-chain alkanes like $C_{20}H_{42}$.

110. (b) Alkanes do not give addition reactions because multiple bond is absent.

111. (a) $\underset{\text{Hexane}}{C_6H_{14}} \xrightarrow[\Delta]{Pyrolysis} \underset{\text{Ethene}}{C_2H_4} + \underset{\text{Butane}}{C_4H_{10}}$

113. (d) In gemdihalide both the halogen atoms are present on the same carbon atom while in vicdihalide both the halogen atoms are present on adjacent carbon atoms.

$\underset{\text{Gemdihalide}}{CH_3-CHBr_2}$ $\qquad$ $\underset{\textit{vic}\text{-dihalide}}{\begin{array}{c} CH_2-CH_2 \\ | \qquad | \\ Br \quad Br \end{array}}$

114. (b) $\underset{\text{2,3,3,3-tetramethylpentane}}{H_3C-H_2C-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3}$

$\underset{\text{isopropyl group 2-methylpentane}}{H_3C-H_2C-H_2C-\overset{\overset{CH_3}{|}}{HC}-CH_3}$

$\underset{\text{2, 2, 3-trimethylpentane}}{H_3C-H_2C-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{HC}}-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3}$ $\quad$ $\underset{\text{3, 3-dimethylpentane}}{H_3C-H_2C-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-CH_3}$

115. **(d)** $CH_4 + C_2H_6 + C_3H_8$ — Natural gas mainly contains methane along with small amounts of ethane and propane.

116. (d) All-butane, Ethane and Propane are possible in this reaction.

117. **(d) All of these — Nitration of propane (vapour phase) gives a mixture of nitroalkanes such as nitromethane, nitroethane, and nitropropane due to free-radical substitution and chain cleavage.**

118. (c) Formation of branches in the chain of *C* atoms

$\underset{\text{straight chain}}{C-C-C-C}$ $\qquad$ $\underset{\text{Branched chain}}{\begin{array}{c} C-C-C \\ | \\ C \end{array}}$

119. (c) Chlorination of alkane in photochomical reaction which takes place by free radical mechanism. Free radicals are formed by homolytic bond fission or homolysis.

120. (c) Marsh gas, Natural gas and coal gas contains CH_4 but producer gas is a mixture of CO and N_2

$$\underset{\text{Coke Red hot}}{2C} + \underbrace{O_2 + 4N_2}_{\text{Air}} \rightarrow \underbrace{2CO + 4N_2}_{\text{Producer gas}}$$

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